Introduction to Mathematical Modeling

Shraddha Puntambekar, Ph.D.

SECTION 1

What is a Mathematical Model?

A simplified, abstract representation of a real-world system using mathematical language

- Translates biological processes into variables, parameters, and equations
- Not a perfect replica — a deliberate simplification that captures essential features
- The goal is not perfection but usefulness for understanding, prediction, and design

> *All models are wrong, but some are useful.*
>
> — George E.P. Box, 1976

Real Biological System → **Abstraction & Simplification** → **Mathematical Model** → **Predictions & Insights**

SECTION 1

Anatomy of a Mathematical Model

Variables

- Quantities that change over time or conditions
- Independent: time (t), space (x)
- Dependent: concentration, population size
- Example: [mRNA](t), [Protein](t), cell count

Parameters

- Fixed quantities we specify or estimate
- Rate constants (k), binding affinities (Kd, Km)
- Degradation rates, diffusion coefficients
- Set the scale and behavior of the model

Equations / Rules

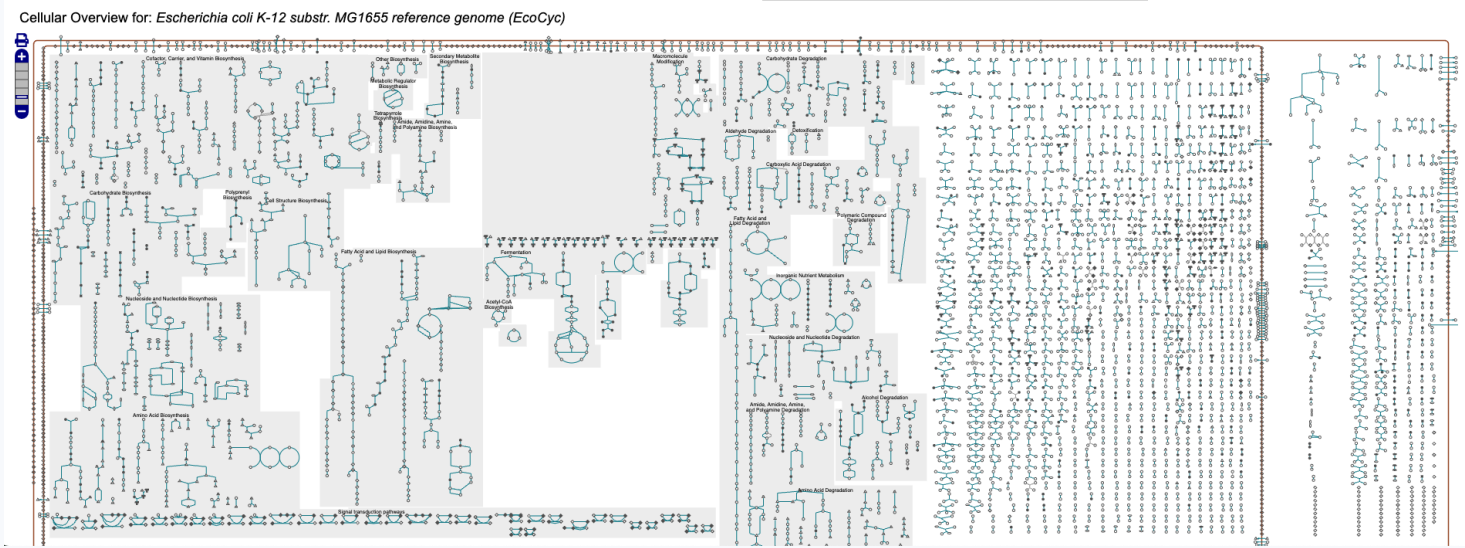
- Relationships connecting variables
- ODE example: $dx/dt = \alpha x - \beta x^2$
- Boolean rule: Gene_ON = TF_1 AND TF_2
- Encode our hypotheses about the system

Initial & Boundary Conditions

- Starting state: concentrations at t = 0
- Spatial constraints: cell membrane, organ edges
- Without these, equations have infinite solutions
- Biologically: what is the system's starting point?

SECTION 1

Biology is astronomically complex!



Genes	4,543
Pathways	478
Enzymatic Reactions	2,419
Transport Reactions	549
Polypeptides	4,494
Protein Complexes	1,210
Enzymes	1,458
Transporters	488
Compounds	3,086
Transcription Units	3,767
tRNAs	89
Growth Media	441
Transcriptional Regulation	6,106
Protein Features	46,219
Phenotype Microarray Datasets	5
GO Annotations	70,767
Gene Essentiality Datasets	6

Task 1: Find similar details for Human (GRCh38) - # of pathways, genes, proteins..

SECTION 1

Simplification is the Point

Biology is astronomically complex

We cannot — and should not — model everything. Modeling requires choosing what matters.

> *The ability to identify what to ignore is the hallmark of good science.*
>
> — Richard Feynman

1 Which components to include?

Genes, proteins, metabolites — not all are relevant to your question

2 Which interactions to represent?

Focus on the regulatory and kinetic relationships that drive the behavior

3 What level of detail?

Genome-scale vs. core pathway? Single-cell vs. population?

The art of modeling = finding the right level of simplification for the question

SECTION 2

Why Mathematical Models in Biology?

Human intuition fails with complexity. When a system has more than ~3 interacting components with nonlinear relationships and feedback, we need mathematics to reason precisely.

Nonlinear Interactions

$x^2 \neq 2x$

Doubling an input rarely doubles the output.

Saturation, cooperativity, and threshold effects are the norm.

Feedback Loops

A → B → A

Positive feedback → switches, bistability.

Negative feedback → oscillations, homeostasis.

Multiple Time Scales

ms → years

Ion channels (ms), signaling (sec), transcription (min), cell division (hrs), evolution (years).

Multiple Spatial Scales

nm → m

Protein folding (nm), organelles (µm), tissues (mm), organs (cm), organisms (m).

Task 2: A -> B -> C and C inhibits A - Can you think how concentrations of each change over time? (no need to simulate now- we will come back to it later)

SECTION 2

The Power of Models: What Can They Do?

1 Prediction

What will happen if we change X?
Drug dosing, gene knockout, treatment schedules, tumor growth, epidemic spread

2 Understanding

Why does the system behave this way?
Mechanism identification, causal reasoning

3 Hypothesis Testing

Is this proposed mechanism consistent with data?
Encode hypothesis → simulate → compare

4 "What If" Scenarios

Cannot ethically test 1000 drug combinations, but can simulate them

5 Design

Engineering biological systems
Synthetic biology, metabolic engineering, drug design

6 Integration

Combine data from different experiments, scales, and -omics modalities

Models serve as the mathematical scaffold that connects hypothesis to data, experiment to theory, and observation to prediction.

SECTION 2

The Design–Build–Test–Learn Cycle

1 DESIGN

Use models to predict optimal designs: pathway configurations, expression levels, gene targets, drug doses

2 BUILD

Construct the biological system: genetic engineering, strain construction, chemical synthesis

Mathematical Model

3 TEST

Measure performance: metabolomics, growth assays, product titers, dose-response

4 LEARN

Compare data to model predictions: update model → refine understanding → iterate

Models are the engine of the Design and Learn steps — they connect prediction to experimental feedback

04

Translating Biology to Mathematics

A Systematic Pipeline — From Biological Observation to Mathematical Expression

1. Problem ID
2. Bio Players
3. Variables & Params
4. Interactions
5. Assumptions
6. Principles
7. Equations
8. Units
9. Well-Posedness

We will walk through each step, then demonstrate all 9 with a complete worked example: the Lac Operon

SECTION 4

Steps 1–3: Question, Players, Variables

1 Define the Research Question

- What phenomenon are you trying to understand, predict, or design?
- The question determines everything: model type, detail level, variables
- Good: "How does E. coli decide to switch from glucose to lactose?"
- Bad: "Model the cell" (too vague)

2 Identify Biological Players

- List all relevant components: genes, mRNAs, proteins, metabolites, ions
- Include structures: compartments, membranes, organelles
- External factors: nutrients, drugs, signals, environment
- Use databases: KEGG, Reactome, UniProt, STRING

3 Define Variables & Parameters

- Variables: quantities that change (what we solve for)
- Parameters: constants we specify (rate constants, Km, Kd)
- Decision depends on question and timescale
- If quantity changes slowly relative to interest → parameter

SECTION 4

Steps 4–6: Interactions, Assumptions, Principles

4 Determine Interactions & Directionality

- How do components affect each other?
- **Activation (+), Inhibition (–), Conversion (→)**
- Directionality matters: A activates B ≠ B activates A
- Represent as a directed graph: Nodes = components, Edges = interactions

5 State Assumptions Explicitly

- **Every model makes assumptions — write them down!**
- Well-mixed compartment (no spatial gradients)
- Quasi-steady state for fast reactions
- Constant temperature, pH, cell volume

6 Identify Governing Principles

Conservation of Mass	Rate of change = production – consumption
Thermodynamic Constraints	$\Delta G < 0$ for spontaneous reactions
Enzyme Kinetics	Michaelis-Menten, Hill, mass action
Stoichiometry	Balanced reactions; atomic conservation

SECTION 4

Steps 7–9: Equations, Units, Well-Posedness

7 Write Mathematical Expressions

Convert each interaction into an equation

For ODE models:
d[X]/dt = (production terms) – (degradation terms)

Choose appropriate kinetic forms:
- Mass action: rate = k · [A] · [B]
- Michaelis-Menten: $V = V_{max} \cdot [S] / (K_m + [S])$
- Hill function: $V = V_{max} \cdot [S]^n / (K^n + [S]^n)$

The form of the equation encodes your biological hypothesis

8 Verify Units

- Every term must have consistent units
- Both sides of equation: same units
- **d[X]/dt → μM/min ⇒ all terms = μM/min**

9 Check Well-Posedness

- Existence: Does a solution exist?
- Uniqueness: Is it the only solution?
- Stability: Robust to small perturbations?
- **Non-negativity: Concentrations ≥ 0!**

SECTION 5

Characterizing Models: Four Axes

Time

Discrete ↔ **Continuous**

Boolean models, difference equations — updates at t, t+1, t+2...

ODEs, PDEs — smooth dynamics in continuous time

State

Discrete ↔ **Continuous**

Boolean (0/1), Petri nets, agent-based with finite states

Concentrations, densities — real-valued variables

Update Rules

Rule-Based ↔ **Equation-Based**

Boolean logic, agent rules

Differential equations, algebraic constraints

Interactions

Abstract ↔ **Mechanistic**

Boolean activation/inhibition, graph-based, statistical

Mass-action kinetics, Michaelis-Menten, thermodynamic

There is no single 'best' combination

SECTION 5

Types of Models

Model Type	Math Basis	Time/State	Strengths	Biological Example
Boolean / Logical	Logic gates (AND/OR/NOT)	Discrete / Discrete	No kinetic params needed; topology-driven	Gene regulatory networks, cell fate decisions
Difference Equations	$x(t+1) = f(x(t))$	Discrete / Continuous	Simple, intuitive; population ecology	Logistic growth, Leslie matrix populations
ODEs	$dx/dt = f(x)$	Continuous / Continuous	Workhorse of systems biology; rich theory	Signaling cascades, gene circuits, PKPD
PDEs	$\partial u/\partial t = D\nabla^2 u + f(u)$	Continuous / Continuous + Space	Spatial patterns; diffusion + reaction	Tumor growth, morphogen gradients, Turing
DDEs	$dx/dt = f(x(t), x(t-\tau))$	Continuous / Continuous	Captures time delays (transcription, transport)	Hematopoiesis, circadian clock
Constraint-Based (FBA)	$S \cdot v = 0$, optimize Z	Steady-state / Continuous	Genome-scale; no kinetic params	E. coli iML1515, yeast GEM

SECTION 5

Additional types of Models

Model Type	Math Basis	Key Feature	When to Use	Biological Example
Stochastic	Master equation; propensity functions	Individual molecular events; intrinsic noise	Low copy numbers (< ~100 molecules)	Gene expression noise, lac stochastic switching
Agent-Based Models (ABM)	Individual agents with rules + environment	Heterogeneous populations; spatial interactions	Tissue-scale; cell heterogeneity	Tumor evolution, immune response, biofilm
PKPD / Compartment	ODE compartments with flow rates	Drug absorption, distribution, metabolism	Pharmacology; drug development	Dosing optimization, virtual clinical trials
Epidemiological (SIR / SEIR)	ODE/stochastic compartments	Population-level disease dynamics	Outbreak prediction; vaccine strategy	COVID-19 models, influenza, malaria
Multi-Scale	Coupled models at multiple scales	Bridges molecular → cellular → tissue	When emergent behavior requires scale coupling	Cardiac (ion → cell → tissue), tumor models

SECTION 5

How to Choose the Right Model

Available Data	Research Question Type: Qualitative Does the system oscillate? How many steady states?	Research Question Type: Quantitative What concentration at what time?
Qual	Boolean/Logical models Network Analysis	Constraint-based (FBA) Stoichiometric models → Steady-state, no kinetics
Quant	ODE/ bifurcation analysis	ODEs / PDEs PKPD models FBA

Real-World Constraints

- Regulatory exposure
  → *Need interpretability? Use mechanistic*
- Available domain expertise
  → *Limited biology knowledge? Use data-driven*
- Computational resources
  → *Limited? Boolean over stochastic*
- Publication / collaboration norms
  → *Field-specific model preferences*

SECTION 6

After Construction: Numerical Simulation

Most biological ODE systems cannot be solved analytically — we rely on numerical methods to compute solutions

Euler Method

Simplest: $x(t+\Delta t) = x(t) + f(x,t)\cdot\Delta t$

✓ Easy to implement and understand

✗ Can be inaccurate; requires small Δt; unstable for stiff systems

Runge-Kutta (RK4/5)

Gold standard: ode45 (MATLAB), solve_ivp (Python scipy)

✓ Accurate, adaptive step size, handles most biological ODEs

✗ Slower than Euler; can still struggle with very stiff systems

Stiff Solvers

Implicit methods: ode15s, BDF, LSODA

✓ Handles multiple timescales (fast + slow reactions together)

✗ More complex; requires Jacobian (analytical or numerical)

SECTION 6

Parameter Estimation & Sensitivity Analysis

Parameter Estimation — Finding the Right Numbers

Least Squares
Minimize $\Sigma(\text{data} - \text{model})^2$. Simple, fast, assumes Gaussian errors.

Maximum Likelihood (MLE)
Find parameters that maximize P(data | model). More general error models.

Bayesian / MCMC
Posterior P(params | data). Full uncertainty quantification. Gold standard.

Approx. Bayesian (ABC)
Simulation-based. No likelihood needed. For complex stochastic models.

Sensitivity Analysis — What Matters Most?

Local Sensitivity
$S = \partial y/\partial p$ • One parameter at a time, near nominal values.
Fast but misses interactions between parameters.

Global Sensitivity (Sobol / Morris)
Varies ALL parameters simultaneously across full ranges.
Sobol indices decompose variance by parameter & interactions.

Common Challenges
- Non-identifiability: different params → same output
- Local minima: optimization gets stuck
- Limited data: under-determined systems
- Curse of dimensionality: many params, little data

SECTION 6

Model Validation: Does It Work?

A model is only as good as its ability to predict data it was NOT trained on — fitting training data is necessary but not sufficient.

Independent Data Validation

Test model predictions against data from experiments NOT used in parameter estimation. The strongest form of validation.

Lac operon model trained on IPTG data → validated on TMG dose-response

Cross-Validation & Holdout

Split data into training and test sets. Train on subset, predict the rest. K-fold cross-validation for robust assessment.

Leave-one-out for small datasets; 80/20 split for larger ones

Predict-Then-Confirm (Prospective)

Use model to make a novel prediction → design experiment to test it → confirm or refute. The gold standard of model validation.

Ozbudak (2004): ODE model predicted bistability → single-cell experiments confirmed

SECTION 7

Where to Find Published Models

BioModels

>2,900 curated models

biomodels.org

Gold standard for published, peer-reviewed models in SBML format. Searchable by pathway, organism, publication.

JWS Online

>900 curated models

jjj.biochem.sun.ac.za

Simulate models directly in the browser — excellent for teaching and exploration. No software install needed.

BioSimulators

>30 simulation tools

biosimulators.org

Aggregates multiple simulation tools under a common standardized interface for reproducible modeling.

Epirecipes

Epidemiological models

epirecipes.org

Repository of SIR/SEIR and related epidemiological models with reproducible code and documentation.

Always check BioModels first when starting a new project — there may already be a published model of your system that you can build on.

SECTION 8

Model Standards & Interoperability

The COMBINE community (Computational Modeling in Biology Network) — international consortium establishing standards for computational biology

SBML

Systems Biology
Markup Language

XML-based standard for representing mathematical models. Defines species, reactions, compartments, parameters, rules. Supported by >300 software tools. Current: Level 3 with FBA & spatial packages.

SED-ML

Simulation Experiment
Description ML

Describes how to simulate a model: which model, which algorithm, what to plot/export. Ensures reproducibility of simulation experiments across tools and labs.

CellML

Component-based
Model Language

Alternative to SBML, developed for the Physiome Project (Peter Hunter, Auckland). Strong in multiscale physiological models. Component-based architecture.

Why Standards Matter: Reproducibility • Sharing • Tool interoperability • Regulatory submission • Always export in SBML when publishing